

Juan David Muñoz-Bolaños

Rechenauerstraße 42, Innsbruck, Austria (Tirol)
+43 676 9115145
juan.munoz@i-med.ac.at

Fraunhofer Institute of Optronics, System Technologies and Image Exploitation (IOSB)

Hiring Team
Gutleuthausstraße 1
76275 Ettlingen, Germany

Subject: Application for Optical Engineer in Adaptive Optics (Requisition 83079)

Dear Dr. Gladysz and Hiring Team,

I am writing to express my strong enthusiasm for the Optical Engineer in Adaptive Optics position (Req. 83079). Concluding my PhD in Innsbruck, my extensive hands-on experience designing and controlling closed-loop adaptive optics hardware directly aligns with Fraunhofer IOSB's research in atmospheric turbulence mitigation and free-space optical communications.

In my current role, I manage the operational readiness of advanced optical systems, implementing AO hardware to compensate for severe biological scattering. The underlying physical principles of wavefront control I utilize are directly applicable to correcting atmospheric distortions in target-in-the-loop systems.

Adaptive Optics & Wavefront Control:

During my PhD, I built multiple closed-loop AO setups from scratch, integrating spatial light modulators, high-power lasers, and sensitive detection hardware. I am well-versed in the precise metrology required to optimize light propagation through complex media.

Software Control & Python Processing:

I have developed high-performance image processing and wavefront control pipelines using Python (and GPU-accelerated JAX). I am eager to apply these skills toward your real-time control and deep learning-based wavefront sensing challenges.

Research & Development Bridge:

From industrial machine vision at INTECOL to prototyping at IPHT Jena and AO research in Innsbruck, I thrive at the interface of theoretical science and practical robustness. I am highly motivated to contribute to Fraunhofer's mission of transforming applied research into critical future technologies.

While my professional language is English, I hold a strong working proficiency in German and am fully capable of writing technical reports. As a Colombian concluding my PhD in Austria, I will be eligible for a graduate work permit, making hiring seamless. I welcome the opportunity to discuss how my AO expertise can support the Adaptive Optics Group.

Sincerely,

Juan David Muñoz-Bolaños

Optical System Engineer
Medical University of Innsbruck

Juan David Muñoz-Bolaños

Optical System Engineer | Adaptive Optics

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PROFESSIONAL SUMMARY

Optical Engineer with hands-on expertise in Adaptive Optics (AO) hardware implementation and wavefront control. Strong background in correcting optical aberrations and light propagation in scattering media, directly applicable to atmospheric turbulence. Proficient in Python-based software control for closed-loop AO systems, image processing, and complex optical metrology.

TECHNICAL COMPETENCIES

Adaptive Optics & Wavefront Control

Hardware design and implementation of closed-loop AO systems. Extensive practical experience with spatial light modulators, wavefront sensors, and aligning complex optical benches for aberration compensation.

Software & Real-Time Control

Programming: Advanced Python (JAX, NumPy) for high-performance data processing pipelines and hardware control. Experience interfacing software with sensitive optical equipment. Embedded: FPGA (LabView) hands-on for real-time applications.

Light Propagation & Metrology

Nonlinear optics and precision metrology focusing on signal-to-noise optimization in photon-starved regimes. Experience handling high-power continuous-wave and pulsed lasers.

PROFESSIONAL EXPERIENCE

Medical University of Innsbruck

Innsbruck, Austria

PhD Researcher - Optical Engineering

Jan 2022 – Present

- **Adaptive Optics Hardware:** Designed, built, and implemented novel closed-loop AO control systems from the ground up to correct severe optical aberrations.
- **Software Control Pipeline:** Developed high-performance, GPU-accelerated Python pipelines to interface with wavefront sensors and control sensitive optical hardware.
- **Complex System Integration:** Prototyped and maintained advanced nonlinear microscopy setups, balancing high-power laser propagation with precise metrology.

Leibniz Institute of Photonic Technology (IPHT)

Jena, Germany

R&D Assistant Researcher

Jun 2020 – Dec 2021

- **System Prototyping:** Assembled vibrational imaging prototypes, integrating fiber probes, spectrometers, and continuous-wave lasers.
- **Data Analysis:** Applied machine learning and structured data analysis techniques to identify complex signal patterns from noisy datasets.

INTECOL S.A.S.

Medellin, Colombia

Machine Vision System Engineer

Mar 2018 – Mar 2019

- **Hardware/Software Interface:** Developed industrial tracking and quality control systems using computer vision, focusing on robust real-time performance.
- **System Reliability:** Managed product lifecycles for safety-critical optical inspection systems, ensuring meticulous operation.

EDUCATION

Medical University of Innsbruck
Ph.D. in Adaptive Optics & Microscopy

Austria
Jan 2022 – Present

Friedrich Schiller University Jena
M.Sc. in Photonics

Germany
2019 – 2021

University of Cauca
B.Sc. in Physics Engineering

Colombia
2012 – 2018

PUBLICATIONS & AWARDS

- **Awards:** ZEISS Summer School (2025), COLFUTURO Grant (2019).
- Muñoz-Bolaños, J. D., et al. Design of a dispersive 1064 nm fiber probe Raman imaging spectrometer and its application to human bladder resectates. *Applied Sciences*, 14(11), 4726 (2024).
- Muñoz-Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* 12(1), 176–184 (2025).
- Sohmen, M., Muñoz-Bolaños, J. D., et al. Optofluidic adaptive optics in multi-photon microscopy. *Biomedical Optics Express* 14(4), 1562–1578 (2023).

LANGUAGES

English (Proficient),
German (Intermediate/Working Proficiency),
Spanish (Native)

REFERENCES

Prof. Monika Ritsch-Marte
Head of Institute for Biomedical Physics
monika.ritsch-marte@i-med.ac.at

Medical University of Innsbruck

Dr. Christoph Krafft
Group leader: Spectroscopy Imaging
christoph.krafft@leibniz-ipht.de

Leibniz Institute of Photonic Technology



URKUNDE

Die Friedrich-Schiller-Universität Jena verleiht durch die
Physikalisch-Astronomische Fakultät
mit dieser Urkunde

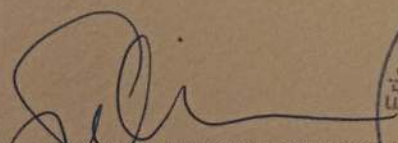
Herrn Juan David Munoz Bolanos
geb. 04.06.1994 in La Cruz (Kolumbien)
den Hochschulgrad

MASTER OF SCIENCE (M.Sc.)

Er hat die Masterprüfung (120 ECTS)
im Studiengang »Photonics«

bestanden.

Jena, 20.12.2021


Prof. Dr. Christian Spielmann
Dekan




Prof. Dr. Silvana Botti
Vorsitzende des
Prüfungsausschusses

Passaporto № / Passport No

MUNOZ BOLANOS

COLOMBIANA

Sexo / Sex
Lugar de nacimiento / Place of birth

Fecha de expedición / Date of issue

Fecha de vencimiento / Date of expiry

G. ANTIOQUIA

Dear Family

P<COLMUNOZ<BOLANOS<<JUAN<DAVID<<<<<<<<<<<<<<<
AV410884<OCOL9406041M2812106CC1061772278<<56

Zertifikat

Muñoz-Bolaños JUAN DAVID

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Schriftliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 16.09.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ö s d

österreich schweiz deutschland

ID-Nummer: ZB2Ö25s34291

Juan Munoz Bolanos

has participated in the

**ZEISS European
Summer School**

Lithography Optics &
Semiconductor Technologies

26 - 27 June 2025

ZEISS Semiconductor Manufacturing Technology, Oberkochen, Germany

Thomas Marschner

Dr. Thomas Marschner
Program Manager Funding Projects

Thomas Stammer

Dr. Thomas Stammer
CTO & Head of Product Strategy



INTECOL

OPTIMIZING THE RHYTHM OF THE WORLD

Medellin, February 4, 2019

TO WHOM IT MAY CONCERN

Integración Tecnológica Industrial Intecol S.A.S. with Tax ID (NIT) 811.041.410-4 certifies that Mr. JUAN DAVID MUÑOZ BOLAÑOS, identified with Citizenship ID No. 1.061.772.278, has been employed by this company since July 3, 2018, under an indefinite term contract, serving in the position of Junior Project Engineer;

earning a salary of One million five hundred thousand pesos (\$1,500,000) plus three hundred thousand pesos (\$300,000) mobility allowance.

I will gladly address any additional information at the phone number 316 3322.

We appreciate your attention,

INTECOL

NIT. 811.041.410-4 - Tel.: 3163822

LENY ADRIANA RUIZ PINO

Administrative and Financial Director

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AGREEMENT FOR GUESTS to carry out a practical training with

Mr. **Juan David Munoz Bolanos** Date of birth: **1994-06-04**
temporarily living at **Emil-Wölk-Str. 7, 07747 Jena** (guest),

Student of the Friedrich-Schiller-University Jena, Abbe School of Photonics, gets in execution of his practical course in the research department Spectroscopy and Imaging, working group Raman and Infrared Histopathology (0104) from 2021-01-01 up to 2021-09-30 admission to the necessary rooms within Leibniz-IPHT.

The use of further devices, media and other infrastructure takes place in agreement with the Heads of the respective departments.

Mr. Juan David Munoz Bolanos has to observe the security regulations for working and other regulations valid in Institute as well as instructions of the laboratory and clean room responsible persons of the Leibniz-IPHT.

Mr. Juan David Munoz Bolanos has to treat all affairs and procedures strictly confidential. Copies, duplicates and others of business and technical information of each kind are forbidden, unless these are sanctioned by the responsible Department Heads.

Publications, reports and lectures about topics which belong together with the fields of work of Leibniz-IPHT require the previous written agreement of the Executive Board.

The liability of Mr. Juan David Munoz Bolanos is limited to cases of wilful action and gross negligence.

Mr. Juan David Munoz Bolanos is insured against accidents on the Friedrich-Schiller-University Jena, Abbe School of Photonics in the context of his/her practical course during his/her length of stay at Leibniz-IPHT.

Jena, 13 November 2020


Prof. J. Popp
Chairman
of Leibniz-IPHT


F. Sondermann
Executive Member
of Leibniz-IPHT


Juan David Munoz Bolanos
Guest